

Enhanced Adversarial Drift Detection for MLOps Feature Monitoring

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Outline

- 1 Motivation & Problem
- 2 Background: C2ST & D3
- 3 EADD Pipeline
- 4 ASPT (Contribution 1)
- 5 SHAP + DSI (Contribution 2)
- 6 Experimental Setup
- 7 Results
- 8 Ablation & Runtime
- 9 Limitations & Future Work
- 10 Conclusion

Motivation: Production ML Models Fail Silently

Distribution drift degrades models in production.

Existing detectors answer only one question:

- **Whether** drift occurred (binary alarm)

Practitioners need to know:

- *Which* features drifted?
- *How severe* is the shift?
- *Is it real* or autocorrelation noise?

D3 [Gözüaık et al., CIKM 2019]

Most robust unsupervised detector (Lukats et al., 2024).

But has **3 gaps**:

- No formal significance test
- No feature attribution
- Linear classifier (misses non-linear drift)

Whether?
Binary alarm

Which?
Feature attribution

How severe?
Severity quantification

No prior work integrates all three in a single unsupervised streaming pipeline.

This paper: Enhanced Adversarial Drift Detection (EADD)
⇒ *whether, when, which, how severe* in one pipeline.

Three Contributions

1 Adaptive Sequential Permutation Test (ASPT)

Gates drift via formal permutation test; **90–95% fewer** permutations vs. fixed budget; preserves Type I error.

2 SHAP-Based Localisation + Drift Severity Index (DSI)

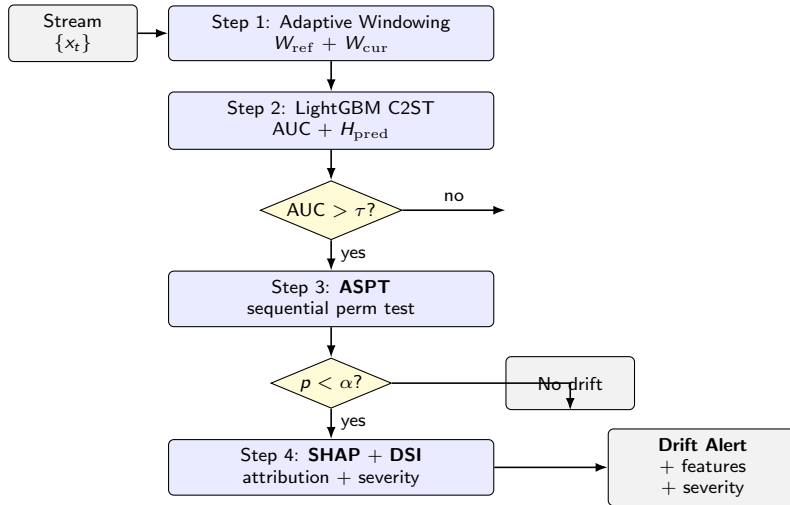
$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s}))$$

Identifies *which* features drifted; quantifies *how severe*.

3 Comprehensive Benchmark

EADD vs. 7 unsupervised detectors on synthetic + 13 real-world streams.

EADD Pipeline



Contribution 1: ASPT

Problem: D3 uses fixed AUC threshold $\rightarrow$ false alarms on autocorrelated data.

Solution: permutation test with adaptive early stopping.

ASPT decision rule (per permutation i)

- **Early reject H_0 :** if $c_i=0$, $i \geq B_{\min}$, $1/(i+1) < \alpha$
 $\rightarrow$ confirm drift with $p = 1/(i+1)$
- **Early accept H_0 :** if $c_i/i > 2\alpha$, $i \geq B_{\min}$
 $\rightarrow$ no drift with $p = (c_i+1)/(i+1)$
- **Terminal:** $p = (c_{B_{\max}}+1)/(B_{\max}+1)$

Result: Drift terminates at $i=20$, stable at $i=10$, vs. $B_{\max}=199$

$\Rightarrow$ **90–95% permutation savings** while preserving overall α .

Contribution 2: SHAP Attribution + DSI

SHAP on the discriminative classifier

TreeSHAP on the LightGBM gate (which D3 discards):

$$\bar{\phi}_j = \frac{1}{n} \sum_i |\phi_j^{(i)}|, \quad s_j = \bar{\phi}_j / \sum_k \bar{\phi}_k$$

⇒ ranks features by drift contribution.

Drift Severity Index (DSI):

$$\text{DSI} = \text{AUC} \times (1 - H_{\text{norm}}(\mathbf{s}))$$

$H_{\text{norm}} \in [0, 1]$ via $\log_2 d$ normalisation →
dimensionality-invariant.

DSI prescription tiers

DSI	Action
> 0.6	Critical: audit pipeline
0.3–0.6	Moderate: shared source
< 0.3	Low: scheduled retrain

(Heuristic tiers; require domain calibration.)

Experimental Setup

Synthetic ($n=5,000$, $d=5$): four drift patterns + four stable streams.

Real-world (13 streams): 4 annotated INSECTS + 9 unannotated.

SHAP attribution ($n=10,000$, $d=10$, drift at $t=5,000$): univariate / subset / multivariate.

Baselines: D3, BNDM, CSDDM, IBDD, OCDD, SPL, UDetect (Lukats et al. 2024).

Configuration: LightGBM (50 trees), $N_{\text{ref}}=500$, $N_{\text{cur}}=200$, $M=50$,
ASPT $B_{\text{max}}=199$, $B_{\text{min}}=10$, $\alpha=0.05$, $\tau=0.7$.

Result 1: Synthetic Coverage + False Alarms

Detector	Types	Coverage	Mean FA	Lift
EADD	4/4	100%	0.00	1.00
BNDM	4/4	100%	4.75	5.75
CSDDM	4/4	100%	5.50	6.50
IBDD	4/4	100%	46.25	47.25
OCDD	4/4	100%	17.50	18.50
D3	2/4	50%	0.00	1.00
SPLL	2/4	50%	0.00	1.00
UDetect	0/4	0%	n/a	n/a

EADD: only detector with full coverage AND zero false alarms.

Result 2: Real-World INSECTS — Faster Detection

Dataset	Prec.	Recall	MTD (EADD)	MTD (D3)
InsectsAbrupt	0.33	0.60	135	155.5
InsectsGradual	n/a	0/1	n/a	n/a
InsectsIncrAbrupt	0.14	1.00	33	272.5
InsectsIncrRecur	0.17	1.00	133	323.5

- EADD beats D3 on MTD in all three jointly detected cases.
- **Up to 8× faster** on incremental drift (33 vs. 272.5).

Result 3: SHAP Attribution Accuracy

Scenario	AUC	P@k	R@k	Top feature (%)	DSI
Univariate (1/10)	0.784	1.0	1.0	F3 (52.3%)	0.193
Subset (3/10)	0.809	1.0	1.0	F5 (24.5%)	0.096
Multivariate (10/10)	0.806	1.0	1.0	F2 (14.6%)	0.016

- Perfect attribution: **P@k = R@k = 1.0** for all 3 scenarios.
- Meaningful result: **univariate** (1/10 features correctly ranked first).
- DSI orders by concentration: univariate > subset > multivariate.

Result 4: False Alarm Robustness

Stable stream	EADD	D3 ($\tau=0.6$)	D3 ($\tau=0.7$)
Gaussian i.i.d.	0	0	0
Autocorrelated AR(1)	0	87.4	54.6
Heteroscedastic	0	10.4	0
Correlated $\rho=0.7$	0	10.0	0

- EADD: **0 false alarms** on all 4 stable types.
- D3 vulnerable to autocorrelation (54.6 FA at $\tau=0.7$).
- Mann–Whitney U: significant at $\tau=0.6$ ($p=0.0101$).

Ablation: Where Does the Gain Come From?

Variant	$d=10$	$d=20$	$d=50$	Coverage
EADD-Full	6.93s	10.76s	23.11s	4/4
EADD _{-ASPT}	6.68s	10.43s	22.64s	4/4
D3	0.07s	0.09s	0.14s	2/4

EADD-Full vs. EADD_{-ASPT} differ by $< 5\%$:

- **Coverage gain** (4/4 vs. D3's 2/4) $\Rightarrow$ **LightGBM upgrade**.
- **ASPT role** $\Rightarrow$ significance control on autocorrelated streams (suppresses 54.6 spurious detections at $\tau=0.7$).
- ASPT's 90–95% saving is *within* the significance stage, orthogonal to LightGBM-driven $\sim 80\text{--}160\times$ overhead vs. D3.

Limitations & Future Work

Limitations

- Wall-clock $\sim 50\text{--}160\times$ D3 — dominated by LightGBM training every $M=50$ steps.
- **Multiple testing over time:** ~ 600 tests over 30K samples; Bonferroni $\alpha' \approx 8.3 \times 10^{-5}$ below ASPT's $1/(B_{\max}+1)=0.005$ floor. Zero-FA result is empirical, not theoretically guaranteed.
- DSI tier boundaries are heuristics, never triggered (all DSI < 0.3 in experiments).
- SHAP attributions are correlational, not causal.
- No isolation of reservoir sampling vs. window config contributions.

Future Work

- Anytime-valid e-values for streaming FWER control.
- Causal attribution methods (Budhathoki et al.).
- Multi-domain DSI calibration to ground tier boundaries.
- Adversarial robustness of discriminative detectors.

EADD extends discriminative drift detection with:

- 1 ASPT + H_{pred} : rigorous confirmation, 90–95% perm savings.
- 2 SHAP + DSI: feature-level diagnosis + severity quantification.
- 3 Benchmark vs. 7 unsupervised detectors on synthetic + real-world.

Headline results:

- 4/4 synthetic coverage with **0** false alarms (only detector to achieve both).
- Up to **8×** faster MTD than D3 on incremental INSECTS streams.
- **P@k = R@k = 1.0** on univariate SHAP attribution.
- **0** false alarms vs. D3's **54.6** on autocorrelated streams ($p=0.0101$).

*EADD answers **whether, when, which, how severe** in one pipeline.*

Thank you.

Questions?

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Code + paper available at the project repository.

Backup: Runtime Scaling — Window N_{cur}

Variant	$N=100$	$N=200$	$N=500$	$N=1000$
EADD-Full	5.83s	6.91s	10.52s	11.22s
EADD- _{ASPT}	5.62s	6.68s	10.23s	10.58s
D3	0.11s	0.07s	0.04s	0.03s

- $N_{\text{cur}}=200 \rightarrow 1000$: EADD time grows only **1.6** $\times$.
- LightGBM cost sub-linear in N , near-linear in d .

Backup: Correlated Features ($d=10$)

Correlation ρ	EADD first-detect	D3 first-detect
0.0	$t=4099$	$t=4117$
0.5	$t=4099$	$t=4117$
0.9	$t=4099$	$t=4117$

- Cholesky-mixed Gaussian; $+2\sigma$ shift in 3 features at $t=4000$.
- EADD detection robust to correlation; stable correlated streams give 0 FA.
- SHAP under correlation $\rightarrow$ interpret at feature-group granularity.